

Fanyi Wu

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EDUCATION

2024-present **University of Manchester**

PhD in Artificial Intelligence

- UKRI fully funded scholarship in collaboration with the University of Cambridge
- Supervised by Dr Michele Caprio and Professor Samuel Kaski on conformal prediction and probabilistic machine learning in collaboration with Google DeepMind

2023-2024 **University of Cambridge**

Mphil in Data Intensive Science

- Thesis: Combinatorial Optimisation – Chaotic Amplitude Control for the Ising Minimisation Using Optical Parametric Oscillator System (Distinction)
- Relevant Modules: Advanced Research Computing, Application of Machine Learning, Deep Learning and Neural Networks, Statistics for Data Science
- Student Representative for the Cohort

2020-2023 **Imperial College, London**

BSci in Physics

- Relevant modules: Computational and experimental Physics, Mechanics and Relativity, Oscillation and Waves, Electrostatic and Magnetism, Statistics, Advanced Mathematics, Thermal Dynamics and Structure of Matters, Quantum Mechanics, Particle Physics, Solid State Physics, Informational Theory
- Full marks for individual Python project in Year 1 Practical Physics Module
- Demonstrated good writing, collaborating, organisational and time-management skills through individual lab reports and group assignments.

PUBLICATION

[1] **F. Wu**, L. Niu, M. Caprio ‘Decoupled Conformal Optimisation: Efficient Prediction Set via Independent Tuning and Calibration,’ ICML EIML Workshop 2026.

[2] L. Cabezas, S. J. Sloman, B. M. Resende, **F. Wu**, M. Caprio, R. Izbicki ‘CREDO: Epistemic-Aware Conformalized Credal Envelopes for Regression,’ arXiv preprint arXiv:2603.06826, 2026.

[3] **F. Wu**, V. Lohmanova, M. Caprio ‘Bayesian Conformal Prediction as Decision Risk Problem Using Bayesian Quadrature,’ NeurIPS EIML Workshop 2025.

[4] A. C. Harwood, S. Vezzoli, T. V. Raziman, C. Hooper, R. Tirole, **F. Wu**, S. Maier, J.B. Pendry, S. A. R. Horsley, R. Sapienza ‘Space-time optical diffraction from synthetic motion,’ Nature Communications 2025

[5] X. Zhang, Q. Wei, Y. Zhu, **F. Wu**, and S. Ananiadou, “THCM-CAL: Temporal-Hierarchical Causal Modelling with Conformal Calibration for Clinical Risk Prediction,” EMNLP 2025.

EXPERIENCE

2026 **University of Manchester – Enhanced xG Modelling for Manchester United**

- Supervising an MSc student in collaboration with Manchester United FC.
- Directed development of a xG framework integrating match features with uncertainty quantification to produce calibrated probabilistic shot-quality estimates.
- Coordinating academic–industry data access with Manchester United, using club-level proprietary tracking datasets for model training and tactical validation.

2025 **Exchange to Aalto University – Optimising Bayesian Conformal Prediction**

- Supervised by Prof Samuel Kaski, fully funded international exchange placement.

- Combined Bayesian posterior predictive density with split conformal prediction and optimise the framework under decision-theoretic view.
- Independently implemented the complete Python codebase and the framework across regression, classification, and out-of-distribution benchmarks.

- 2024 University of Cambridge MPhil Thesis – Chaotic Amplitude Control for the Ising Minimization Using Optical Parametric Oscillator System**
- Supervised by Prof Natalia Berloff in Department of Mathematics
 - Implemented an algorithm for analog bistable systems to solve combinatorial optimization problems (MAX-CUT and GSets)
 - Optimized the algorithm using Machine Learning techniques to efficiently find low-energy states of the Ising Hamiltonian, demonstrating performance comparable to state-of-the-art heuristics.
- 2024 University of Cambridge – Degradation Strategies in Diffusion Model**
- Applied novel degradation strategies within DDPMs on the MNIST dataset.
 - Using a CNN to explore hyperparameter impacts of Diffusion Model.
 - Developed and tested Gaussian blurring and "crop-flip" degradation strategies.
 - Evaluated models using Inception scores, FID, and classification confidence.
- 2023 Imperial College London – Quantum Optics & Meta-material Group**
- Supervised by Prof Riccardo Sapienza.
 - Explored space-time double-slit experiment using a time-varying meta-material ITO.
 - Conducted data analysis and simulations, showed interference patterns in frequency space
 - Wrote arbitrary patterns and produced computer generated holograms (CGH) on spatial-light modulator.
- 2022 Imperial College London BSc Project – Single Photon Pairs Generation and Detection and Building a Micro-spectrophotometer**
- Supervised by Dr Stefano Vezzoli to explore SPDC behaviour of photons with EMCCD.
 - Conducted experiment on optical behaviour of BBO non-linear crystals.
 - Used thin films of meta surface BBO to trap photons and simulate SPDC behaviour by modifying the codes and compare with experimental results.
- 2022 Imperial College London – Magnetic Confinement Fusion**
- Studied plasma fluid & kinetic models and plasma physics in tokamak.
 - Learned command and terminal basics to control computer using Linux.
 - Used SOL-KiT to do large data simulations on SOL plasma behaviour.
 - Process large data set on HPC and managed to do data analysis with the generated results.
- 2021 Zhejiang University High-Energy & Particle Physics Research Programme**
- Learned Cpp fundamentals and used ROOT to extrapolate data and explore the underlying relationships between variables.
 - Broadened awareness of theories in High-Energy & Particle Physics and relevant fields.

CONFERENCES & WORKSHOPS

2026 Poster presentation, 2nd Workshop on Epistemic Intelligence on Machine Learning (EIML) at Forty-Third International Conference on Machine Learning (ICML), Seoul Korea.

2026 Poster presentation, The 18th World Meeting of the International Society for Bayesian Analysis (ISBA), Nagoya, Japan.

2026 Selected talk speaker and poster presentation, Workshop on Uncertainty in Machine Learning (WUML), Tartu, Estonia.

2025 Poster presentation, 1st Workshop on Epistemic Intelligence on Machine Learning (EIML) at European NeurIPS (EurIPS), Copenhagen, Denmark.

2025 Poster presentation, Royal Statistical Society (RSS) International Conference, Edinburgh, UK.

2025 Poster presentation and attendee, Cambridge ELLIS Unit Summer School on Probabilistic Machine, Cambridge, UK.

2025 Poster presentation, Post-Bayesian Workshop, University College London (UCL), London, UK.

RESEARCH VISITS & INTERNS

Visiting Scholar

Mar. 2026

Department of Informatics,
King's College London, London, United Kingdom
Hosting Professor: Nicola Paoletti

Visiting Scholar

Mar. 2025 – Apr. 2025

Department of Computer Science,
Aalto University, Espoo, Finland
Hosting Professor: Samuel Kaski

Visiting Scholar

Jun. 2024 – Sept. 2024

Department of Applied Mathematics,
University of Cambridge, Cambridge, United Kingdom
Hosting Professor: Natalia Berloff

Research Assistant

Jul. 2023 – Sept. 2023

Department of Physics,
Complex Nano-photonics Group,
Imperial College London, London, United Kingdom
Hosting Professor: Riccardo Sapienza

Research Assistant

Jun. 2022 – Aug. 2022

Department of Physics,
Plasma Group,
Imperial College London, London, United Kingdom
Hosting Professor: Robert Kingham

Research Assistant

Jun. 2021 – Aug. 2021

Department of Physics,
High-Energy and Particle Physics Group,
Zhejiang University, Hangzhou, China
Hosting Professor: Hongbo Zhu

ADDITIONAL SKILLS AND ACHIEVEMENTS

IT SKILLS

- Python, MATLAB, Linux, C++, Machine Learning
- MS Word, PowerPoint, Excel

LANGUAGES

- Chinese, English

ACHIEVEMENTS

- 2026 Junior Researcher Award for International Society for Bayesian Analysis
- 2025 Turing scheme award winner for fully funded international exchange
- 2024 CVPR Embodied AI Workshop PRS Challenge: Ranked top2 in this challenge

- 2019 BPhO: Global Gold Prize, ranking top 10 in China
- 2019 21th Century Speech Competition: Outstanding Winner
- 2018 ASDAN 3D Printing Competition: National Championship